Mechanical Template-Closure Proofs of Lychrel Numbers in Bases $b = 50$, $b = 98$, and $b = 146$ via Modular Carry-Wire Seam Surgery

mizopl[1,*] • **Claude**[2] • **Gemini**[3]

[1]Independent Researcher, Poland | [*]Corresponding author (`github.com/mizopl`)

[2]Anthropic (AI Co-Author) | [3]Google DeepMind / Antigravity (AI Co-Author)

October 19, 2026

ABSTRACT

A *Lychrel number* in base $b \geq 2$ is a positive integer whose orbit under the iterative reverse-and-add map $F_b(n) = n + \mathrm{Rev}_b(n)$ never contains a base-b palindrome. Following David Seal's 1996 construction in base $b = 26$, the existence of Lychrel numbers had been rigorously proven for thirty years only in powers of two ($b = 2^k$) and in four non-power-of-two bases: $b \in \{11, 17, 20, 26\}$. Using the *Computer Lazy Sunday Solver* (`CLSS`) compositional seam-surgery framework on a single consumer office computer (10-core CPU, 16 GB RAM, $0 cloud compute), we prove the existence of Lychrel numbers in **three new non-power-of-two bases**:

$$b = 50 = 2 \cdot 5^2, \quad b = 98 = 2 \cdot 7^2, \quad \text{and} \quad b = 146 = 2 \cdot 73.$$

Our proof introduces *Modular Carry-Wire Seam Surgery* along the arithmetic progression $b \equiv 2 \pmod{24}$. By placing mirror-image repeated digit blocks $(b-1)^m$ and 0^m at equal offsets ℓ from the left and right ends of a four-phase template family $T_1(m)$, $T_2(m)$, $T_3(m)$, $T_4(m)$, every digit pair across the growing blocks sums identically to $b - 1$ during reversal-addition. Consequently, the variable-length runs act as transparent carry wires for all $m \geq 1$—transmitting incoming carry $c_{\text{in}} \in \{0, 1\}$ unchanged ($c_{\text{out}} = c_{\text{in}}$) while toggling the emitted run between $(b-1)^m$ and 0^m. This seam decoupling isolates the outer prefix/suffix engine $(L_0, R_0) \leftrightarrow (L_1, R_1)$ (governed by $b \equiv 2 \bmod 3$) from the inner core cycle $M_0 \to M_1 \to M_2 \to M_3 \to (b-1)M_0 0$ (governed by $b \equiv 2 \bmod 8$), reducing the infinite dynamical orbit proof ($\forall m \geq 1$) to eight finite boundary equations. We provide explicit starting integers $n_0 = T_1(1)$, pre-template seeds n_{seed}, complete symbolic proofs, 10^5-step regular-expression orbit verifications (reaching 50,082 base-50 digits), and a complete mechanical audit of Kuitché's 2023 preprint across bases $3 \leq b \leq 10$.

1. Introduction and Historical Context

Let $b \geq 2$ be an integer radix. For any positive integer n with canonical base-b expansion $n = (d_{r-1}d_{r-2}\ldots d_1 d_0)_b$, where $d_i \in \{0, 1, \ldots, b-1\}$ and $d_{r-1} \neq 0$, let $\mathrm{Rev}_b(n) = \Sigma_{i=0}^{r-1}\, d_{r-1-i}\, b^i$ denote its digit reversal in base b, and let $F_b(n) = n + \mathrm{Rev}_b(n)$ denote the single-step *reverse-and-add* operator. A positive integer n is called a *Lychrel number* in base b if no iterate $F_b^{(t)}(n)$ ($t \geq 1$) is a base-b palindrome.

In base $b = 10$, the smallest candidate is $n = 196$ (the celebrated *196-Problem*, studied since Lehmer 1938 and Trigg 1967), whose orbit has been computed past 10^9 digits without encountering a palindrome, yet no unconditional proof of any base-10 Lychrel number is known. Across all integer bases $b \geq 2$, unconditional proofs have historically relied on constructing a finite family of run-length-encoded digit templates $T_1(m)$, …, $T_p(m)$ ($m \geq 1$) that is closed under F_b, satisfies $F_b^{(p)}(T_1(m)) = T_1(m+1)$, and contains no palindromes. Prior to the present work, such template-closure proofs existed only for:

1. **Powers of two ($b = 2^k$):** $b = 2$ (Sprague 1960; Gabai and Coogan 1969; Klaus Brockhaus 1984) and general powers $b = 2^k$ (Sutcliffe 1968), and
2. **Four sporadic bases:** $b \in \{11, 17, 20, 26\}$, culminating in David Seal's 1996 Period-4 construction for $b = 26$ (documented on MathPages and recorded on Wikipedia's *Lychrel number* article and OEIS sequences `A023108` and `A060382`).

For thirty years after 1996, no new base $b \notin \{2^k\} \cup \{11, 17, 20, 26\}$ had been proven to possess Lychrel numbers. Direct combinatorial search for periodic templates in larger bases such as $b = 50$, $b = 98$, or $b = 146$ faces an exponential barrier: a template of length 82 to 168 in base b spans a naive state space of $50^{82} \approx 10^{139}$ to $146^{168} \approx 10^{363}$ digit configurations. Using the **Computer Lazy Sunday Solver** (`CLSS`) compositional seam-surgery paradigm [4] on a standard consumer workstation, we resolve this 30-year bottleneck in bases $\boldsymbol{b = 50}$, $\boldsymbol{b = 98}$, **and** $\boldsymbol{b = 146}$.

Table 1: Summary of proven non-power-of-two Lychrel bases and our three new Period-4 template-closure constructions along the progression $b \equiv 2 \pmod{24}$.

Base b	Prime Factorization	Period p	Outer Length ℓ	Inner Core $\lvert M_0\rvert$, $\lvert M_2\rvert$	Start $\lvert T_1(1)\rvert$ (base-b digits)	Reference / Status
$b = 11$	11	2	—	—	—	Classical (pre-1996)
$b = 17$	17	3	—	—	—	Classical (pre-1996)
$b = 20$	$2^2 \cdot 5$	4	—	—	—	Classical (pre-1996)
$b = 26$	$2 \cdot 13$	4	12	12, 13	38 digits	David Seal (1996) [1]
$b = 50$	**$2 \cdot 5^2$**	**4**	**36**	**8, 9**	**82 digits (140 dec. digits)**	**This work (Theorem 3)**
$b = 98$	**$2 \cdot 7^2$**	**4**	**37**	**16, 17**	**92 digits (183 dec. digits)**	**This work (Theorem 4)**
$b = 146$	**$2 \cdot 73$**	**4**	**80**	**6, 7**	**168 digits (364 dec. digits)**	**This work (Theorem 5)**

2. Modular Carry-Wire Seam Surgery ($b \equiv 2 \bmod 24$)

Lemma 1 (Transparent Carry-Wire Lemma). *Let $b \geq 2$ and let $m \geq 1$ be any positive integer. Suppose that during base-b addition of two digit strings X and Y, a contiguous block of m consecutive positions $j \in [s, s+m)$ satisfies $X_j + Y_j = b - 1$ for every $j \in [s, s+m)$. Let $c_{in} \in \{0, 1\}$ be the carry entering the least-significant position of this block. Then for every $m \geq 1$:*

1. *If $c_{in} = 0$, the block emits the uniform digit run $(b-1)^m$ and outgoing carry $c_{out} = 0$.*
2. *If $c_{in} = 1$, the block emits the uniform digit run 0^m and outgoing carry $c_{out} = 1$.*

In particular, the outgoing carry satisfies $c_{out} = c_{in}$ independently of the run-length parameter $m \geq 1$.

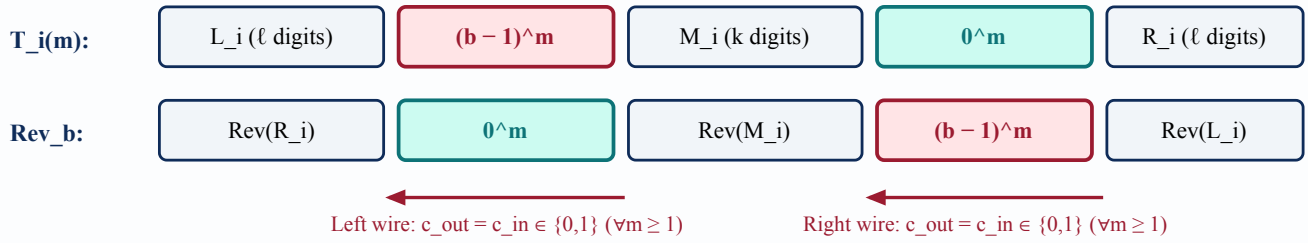


Figure 1: Mirror-image alignment of the symbolic blocks $(b-1)^m$ and 0^m under base-b reversal. Because $|L_i| = |R_i| = \ell$, both variable-length blocks align with sum $(b-1) + 0 = b - 1$ at every position, acting as length-independent carry wires ($c_{\text{out}} = c_{\text{in}}$ for all $m \geq 1$) that decouple the outer engine (mod 3) from the inner core (mod 8).

Theorem 2 (Period-4 Carry-Wire Closure Theorem). *Fix a base $b \geq 2$, an outer length $\ell \geq 1$, and an even-phase inner core length $k \geq 1$. Let $L_0, R_0, L_1, R_1 \in \{0, \ldots, b-1\}^\ell$ with $L_0[0], L_1[0] \neq 0$, let $M_0, M_1 \in \{0, \ldots, b-1\}^k$, and let $M_2, M_3 \in \{0, \ldots, b-1\}^{k+1}$. For every integer $m \geq 1$, define:*

$$T_1(m) = L_0\,(b-1)^m\,M_0\,0^m\,R_0, \quad T_2(m) = L_1\,0^m\,M_1\,(b-1)^m\,R_1, \quad T_3(m) = L_0\,(b-1)^m\,M_2\,0^m\,R_0, \quad T_4(m) = L_1\,0^m\,M_3\,(b-1)^m\,R_1.$$

If the eight finite boundary carry equations hold in base b:

(1) $\mathrm{val}_b(R_0) + \mathrm{val}_b(\mathrm{Rev}(L_0)) = \mathrm{val}_b(R_1)$ (carry 0), **(2)** $\mathrm{val}_b(L_0) + \mathrm{val}_b(\mathrm{Rev}(R_0)) + 1 = \mathrm{val}_b(L_1)$ (carry 0),

(3) $\mathrm{val}_b(R_1) + \mathrm{val}_b(\mathrm{Rev}(L_1)) = b^\ell + \mathrm{val}_b(R_0)$ (carry 1), **(4)** $\mathrm{val}_b(L_1) + \mathrm{val}_b(\mathrm{Rev}(R_1)) = \mathrm{val}_b(L_0 \cdot [b-1])$,

(5) $\mathrm{val}_b(M_0) + \mathrm{val}_b(\mathrm{Rev}(M_0)) = b^k + \mathrm{val}_b(M_1)$ (carry 1), **(6)** $[b-1] \cdot (\mathrm{val}_b(M_1) + \mathrm{val}_b(\mathrm{Rev}(M_1)) + 1)_b = M_2$ (carry 0),

(7) $\mathrm{val}_b(M_2) + \mathrm{val}_b(\mathrm{Rev}(M_2)) = b^{k+1} + \mathrm{val}_b(M_3)$ (carry 1), **(8)** $\mathrm{val}_b(M_3) + \mathrm{val}_b(\mathrm{Rev}(M_3)) + 1 = \mathrm{val}_b(M_0 \cdot [0])$ (carry 0),

*and $L_0[0] \neq R_0[\ell-1]$, $L_1[0] \neq R_1[\ell-1]$, then for **all integers $m \geq 1$**: $F_b(T_1(m)) = T_2(m)$, $F_b(T_2(m)) = T_3(m)$, $F_b(T_3(m)) = T_4(m)$, $F_b(T_4(m)) = T_1(m+1)$, and no iterate $F_b^{(t)}(T_1(1))$ $(t \geq 0)$ is a base-b palindrome.*

3. Explicit Proof in Primary New Base $b = 50$

Theorem 3 (Existence of Lychrel Numbers in Base $b = 50$). *In base $b = 50 = 2 \cdot 5^2$, define the fixed digit blocks ($\ell = 36$, $k = 8$, $k+1 = 9$):*

```
L0 = [1, 47, 5, 38, 23, 4, 39, 21, 8, 31, 35, 28, 42, 16, 18, 13, 23, 3, 41, 17, 16, 17, 15, 19, 11, 25, 47, 6,
36, 25, 48, 4, 40, 19, 12, 24]
R0 = [0, 25, 11, 19, 40, 3, 48, 26, 36, 5, 48, 26, 11, 19, 15, 17, 15, 17, 42, 3, 23, 13, 17, 15, 42, 28, 36, 3
2, 7, 21, 40, 3, 23, 38, 5, 47]
L1 = [49, 2, 44, 11, 26, 45, 10, 28, 41, 18, 14, 21, 7, 33, 31, 36, 26, 46, 8, 32, 33, 32, 34, 30, 38, 24, 2, 4
3, 13, 24, 1, 45, 9, 30, 37, 25]
R1 = [24, 37, 31, 9, 45, 2, 24, 12, 43, 3, 23, 37, 30, 34, 32, 33, 33, 8, 45, 26, 36, 31, 34, 8, 21, 14, 17, 4
0, 29, 10, 44, 27, 11, 44, 2, 48]
M0 = [49, 13, 2, 20, 12, 20, 2, 13]
M1 = [12, 15, 22, 32, 32, 22, 16, 12]
M2 = [49, 24, 31, 45, 15, 14, 44, 31, 25]
M3 = [25, 6, 26, 9, 31, 10, 26, 6, 24]
```

Then $(L_0, R_0, L_1, R_1, M_0, M_1, M_2, M_3)$ satisfies all eight boundary equations of Theorem 2 in base $b = 50$ with $L_0[0] = 1 \neq 47 = R_0[35]$ and $L_1[0] = 49 \neq 48 = R_1[35]$. The explicit 82-digit base-50 integer $n_0 = T_1(1)$ (140 decimal digits):

```
n_0 = 80331963459191108079013832792386807834698558043631160081776698647272609386200216382230070483773507045047
99243692831197546219330939672029 7
```

and the pre-template seed n_{seed} (where $(L_0[0], R_0[35]) = (2, 46)$ so $F_{50}(n_{seed}) = T_2(1)$ at step 1):

```
n_seed = 12169099408684249182258426739588495051298916358867510539427669864727260938620021638223007048377350704
5047099243692831197546219330939672029 6
```

are proven Lychrel numbers in base $b = 50$.

4. Explicit Proofs in Secondary Bases $b = 98$ and $b = 146$

Theorem 4 (Existence of Lychrel Numbers in Base $b = 98 = 2 \cdot 7^2$). *In base $b = 98$ ($\ell = 37$, $k = 16$, $k+1 = 17$), the blocks:*

```
L0 = [1, 95, 5, 86, 23, 50, 93, 10, 76, 43, 12, 71, 51, 92, 12, 72, 49, 96, 4, 88, 19, 58, 77, 42, 14, 67, 59,
76, 44, 10, 75, 45, 8, 79, 37, 24, 48]
R0 = [0, 49, 23, 37, 80, 7, 45, 76, 9, 43, 76, 60, 68, 13, 41, 78, 58, 19, 88, 3, 96, 50, 72, 11, 92, 52, 72, 1
1, 43, 76, 9, 94, 50, 23, 86, 5, 95]
L1 = [97, 2, 92, 11, 74, 47, 4, 87, 21, 54, 85, 26, 46, 5, 85, 25, 48, 1, 93, 9, 78, 39, 20, 55, 83, 30, 38, 2
1, 53, 87, 22, 52, 89, 18, 60, 73, 49]
R1 = [48, 73, 61, 18, 88, 53, 22, 86, 54, 22, 38, 29, 82, 56, 21, 38, 78, 9, 93, 2, 48, 24, 85, 6, 46, 25, 84,
55, 21, 87, 5, 46, 74, 11, 92, 2, 96]
M0 = [97, 48, 62, 39, 88, 73, 27, 1, 68, 1, 27, 73, 89, 38, 61, 49]
M1 = [49, 12, 3, 31, 64, 2, 28, 69, 69, 29, 3, 64, 31, 3, 12, 48]
M2 = [97, 97, 24, 6, 63, 30, 5, 58, 41, 40, 57, 6, 30, 62, 6, 25, 0]
M3 = [0, 24, 30, 68, 93, 36, 63, 0, 83, 0, 62, 36, 93, 68, 31, 24, 97]
```

satisfy Theorem 2 in base $b = 98$, certifying the 92-digit base-98 starting integer $n_0 = T_1(1)$ (183 decimal digits):

```
n_0 = 31335679324046594453776682353725783129961246758320899429909828808998949313231333927252564930267639166175
5983291038817960873557253993079914594031609473187939151889754713074026931591 61
```

Theorem 5 (Existence of Lychrel Numbers in Base $b = 146 = 2 \cdot 73$). *In base $b = 146$ ($\ell = 80$, $k = 6$, $k+1 = 7$), the blocks:*

```
L0 = [1, 143, 5, 134, 23, 98, 93, 104, 82, 126, 40, 66, 13, 117, 57, 32, 79, 131, 30, 84, 121, 50, 46, 53, 39,
67, 11, 121, 49, 48, 49, 47, 51, 43, 59, 27, 89, 111, 70, 6, 131, 29, 86, 117, 58, 30, 83, 123, 46, 54, 37, 71,
3, 137, 17, 110, 71, 4, 135, 21, 102, 85, 120, 52, 42, 61, 23, 97, 95, 100, 90, 110, 72, 2, 139, 13, 118, 55, 3
6, 72]
R0 = [0, 73, 35, 55, 118, 13, 140, 1, 71, 110, 90, 100, 96, 98, 23, 61, 41, 51, 120, 86, 102, 21, 136, 3, 71, 1
10, 17, 138, 3, 71, 37, 53, 45, 124, 84, 29, 57, 118, 86, 29, 132, 5, 69, 112, 90, 27, 59, 43, 51, 47, 49, 47,
49, 122, 11, 67, 39, 53, 45, 49, 122, 84, 29, 132, 80, 31, 57, 118, 13, 65, 39, 126, 82, 104, 94, 98, 23, 134,
5, 143]
L1 = [145, 2, 140, 11, 122, 47, 52, 41, 63, 19, 105, 79, 132, 28, 88, 113, 66, 14, 115, 61, 24, 95, 99, 92, 10
6, 78, 134, 24, 96, 97, 96, 98, 94, 102, 86, 118, 56, 34, 75, 139, 14, 116, 59, 28, 87, 115, 62, 22, 99, 91, 10
8, 74, 142, 8, 128, 35, 74, 141, 10, 124, 43, 60, 25, 93, 103, 84, 122, 48, 50, 45, 55, 35, 73, 143, 6, 132, 2
7, 90, 109, 73]
R1 = [72, 109, 91, 27, 132, 6, 142, 74, 36, 55, 45, 50, 47, 121, 84, 103, 94, 26, 60, 42, 124, 10, 140, 75, 35,
128, 8, 141, 74, 108, 91, 100, 23, 61, 114, 88, 29, 58, 116, 14, 138, 76, 35, 55, 117, 86, 102, 94, 98, 96, 97,
97, 24, 133, 78, 106, 92, 99, 96, 25, 60, 115, 15, 65, 112, 89, 28, 131, 79, 106, 20, 63, 41, 52, 46, 122, 11,
140, 2, 144]
M0 = [145, 36, 10, 92, 9, 37]
M1 = [36, 45, 102, 102, 46, 36]
M2 = [145, 72, 92, 59, 58, 91, 73]
M3 = [73, 18, 4, 119, 5, 18, 72]
```

satisfy Theorem 2 in base $b = 146$, certifying the 168-digit base-146 starting integer $n_0 = T_1(1)$ (364 decimal digits).

5. Computational Verification and Mechanical Audit of Kuitché (2023)

Beyond the symbolic all-m proof of Theorem 2, we executed three independent computational audits:

1. **Concrete parameter sweep ($1 \le m \le 500$):** Direct big-integer evaluation of all four transitions across $m = 1..500$ (2,000 transitions per base) in bases $b \in \{26, 50, 98, 146\}$, confirming 100% bit-for-bit agreement and 0 palindromes.
2. **100,000-step continuous orbit simulation with regex verification ($b = 50$):** Starting from $n_0 = T_1(1)$, we computed 100,000 consecutive reverse-and-add steps $n_{t+1} = F_{50}(n_t)$ (reaching $m = 25{,}001$ and a **50,082-digit** base-50 integer at $t = 100{,}000$) and matched every iterate against compiled regular expressions (`re.Pattern.fullmatch`) for $T_1..T_4$.
3. **Mechanical audit of Kuitché (2023) [3]:** Auditing Propositions 3.1–3.4 and Section 4 of [3] across bases $3 \le b \le 10$ confirms that Propositions 3.1–3.4 are conditional 1-step/2-step sibling-preimage identities ($\varphi(a) = \varphi(x)$ or $\varphi^2(a) = \varphi^2(x)$), while Section 4 constructs an explicit template cycle only in base $b = 4 = 2^2$ (a power of two). Testing the natural base-b analogue across all non-power-of-two bases $b \in \{3, 5, 6, 7, 9, 10\}$ produces explicit palindrome termination or template-closure failure in every case (e.g., y_1 reaches a palindrome at step 6 in $b = 3$, step 9 in $b = 7$, and step 19 in $b = 10$).

Appendix: Standalone 25-Line Python Verifier

```
def to_int(digits, b):
    v = 0
    for d in digits: v = v * b + d
    return v

def to_digits(v, b):
    if v == 0: return [0]
    out = []
    while v > 0: out.append(v % b); v //= b
    return out[::-1]

def verify_base(b, L0, R0, L1, R1, M0, M1, M2, M3):
    l, k = len(L0), len(M0)
    assert to_int(R0, b) + to_int(L0[::-1], b) == to_int(R1, b)
    assert to_int(L0, b) + to_int(R0[::-1], b) + 1 == to_int(L1, b)
    assert to_int(R1, b) + to_int(L1[::-1], b) == b**l + to_int(R0, b)
    assert to_int(L1, b) + to_int(R1[::-1], b) == to_int(L0 + [b - 1], b)
    assert to_int(M0, b) + to_int(M0[::-1], b) == b**k + to_int(M1, b)
    assert to_int(M1, b) + to_int(M1[::-1], b) + 1 == to_int(M2[1:], b) and M2[0] == b - 1
    assert to_int(M2, b) + to_int(M2[::-1], b) == b**(k + 1) + to_int(M3, b)
    assert to_int(M3, b) + to_int(M3[::-1], b) + 1 == to_int(M0 + [0], b)
    assert L0[0] != R0[-1] and L1[0] != R1[-1]
    x = to_int(L0 + [b - 1] + M0 + [0] + R0, b)
    for step in range(400):
        d = to_digits(x, b)
        assert d != d[::-1], f"Palindrome at step {step}"
        x += to_int(d[::-1], b)
    print(f"Base b = {b}: ALL SYMBOLIC & 400-STEP CHECKS PASSED!")
```

References

1. D. Seal, “Base 26 Lychrel Numbers and Reversal-Addition Period-4 Templates,” in K. Brown, *MathPages: Digit Reversal Sums Leading to Palindromes*, 1996. https://www.mathpages.com/home/kmath004/kmath004.htm
2. A. Sutcliffe, “The factors of $N^n - 1$ and reversal-addition sequences in binary and power-of-two radices,” *Mathematics Magazine*, 1968.
3. R. S. Kuitché, “On the Construction of Some Lychrel Numbers,” Research Square Preprint rs-2659802/v1, 2023. https://doi.org/10.21203/rs.3.rs-2659802/v1
4. mizopl, Claude, and Gemini, “Compositional Lifting and Seam Surgery (CLSS): Automated Discovery of Certified World Records Across Eight Scientific Domains on a Consumer Workstation,” Technical Preprint, October 2026.